

DOEF

让负荷预测服务于 储能削峰决策

面向储能决策的集成负荷预测方法

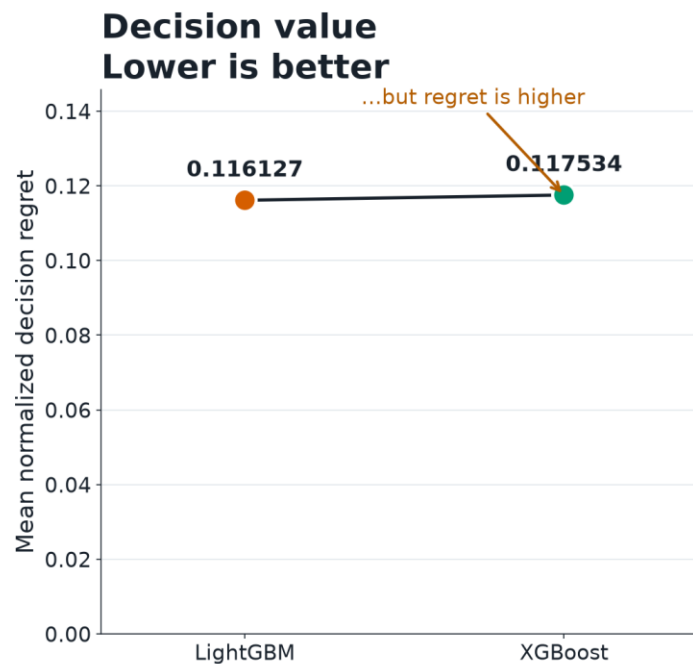
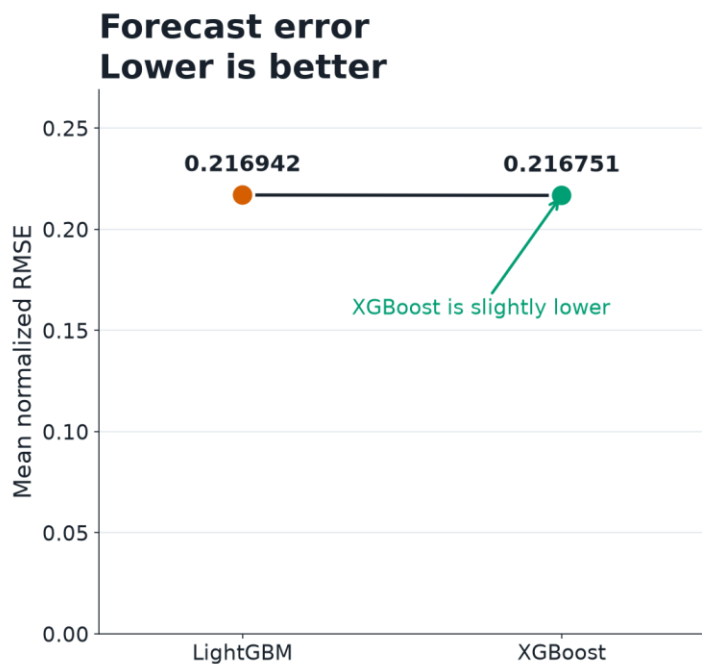


预测精度与削峰效果的排序差异

八建筑测试结果中，XGBoost 的 RMSE 略低于 LightGBM，决策后悔值却略高。

Forecast metrics alone do not determine decision value

Frozen eight-building Test aggregate · equal-building mean after Train-mean normalization



Directional counterexample only: the gap is small and project-scoped; it does not imply that forecast accuracy is unimportant.

指标排序差异

RMSE

XGBoost
0.216751

Decision regret

LightGBM
0.116127

预测指标与调度指标
需要分别评价

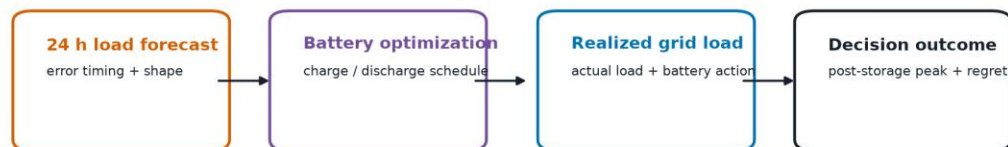
预测精度仍是必要指标

负荷预测驱动储能调度

预测生成充放电计划，再用实际负荷计算调度后的取电峰值。

Prediction is an input to the decision — not the endpoint

The same scalar forecast score can hide different errors at decision-critical hours



Evaluate both forecast quality and downstream storage value

CONCEPTUAL / EXPLANATORY · no experimental values encoded · no cost, carbon, or energy-savings claim

流程示意

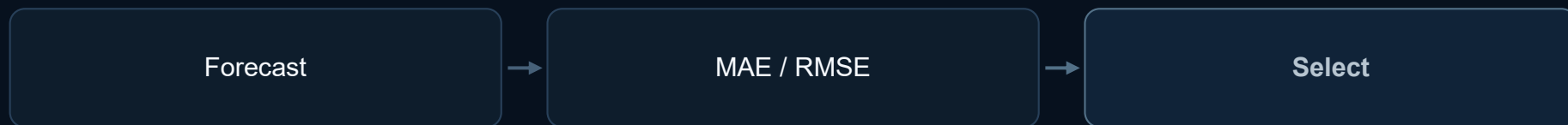
各预测方法共享电池约束和优化器，比较预测输入对调度结果的影响。

Conceptual asset: outputs/phase13_1/presentation_assets/con_01_prediction_decision_bridge.png

以验证集削峰表现选择融合权重

预测分量与电池约束保持一致，融合权重依据验证集决策后悔值选择。

传统预测选择



预测误差最小

DOEF 选择



决策 regret 最小

权重选择依据：验证集决策后悔值

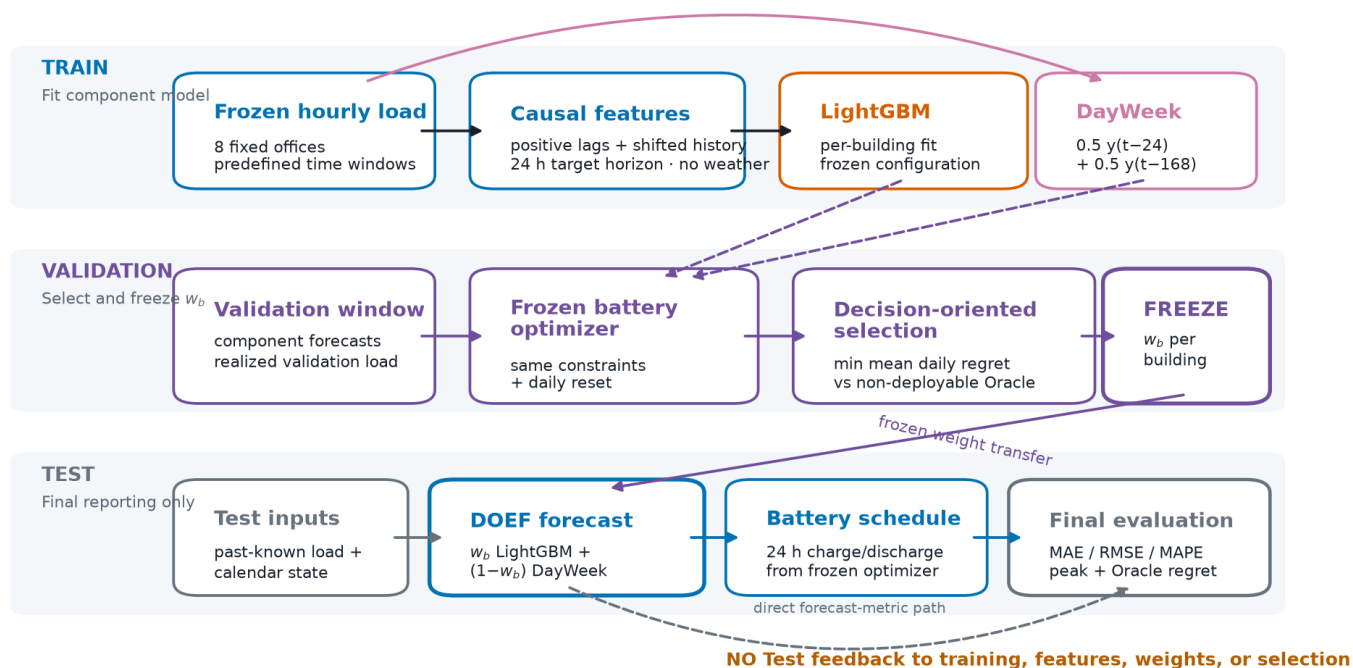
预测器与优化器分别运行，权重在验证集确定

DOEF: 周期基线与 LightGBM 融合

逐建筑选择 DayWeek 与 LightGBM 的权重，验证完成后固定权重并评价测试集。

DOEF v1.0 — frozen causal architecture

Prediction serves the storage decision · Validation selects · Test evaluates only



ALGORITHM FROZEN · deterministic ensemble and constrained optimization · not joint training, bilevel optimization, online learning, or a deployable Oracle.

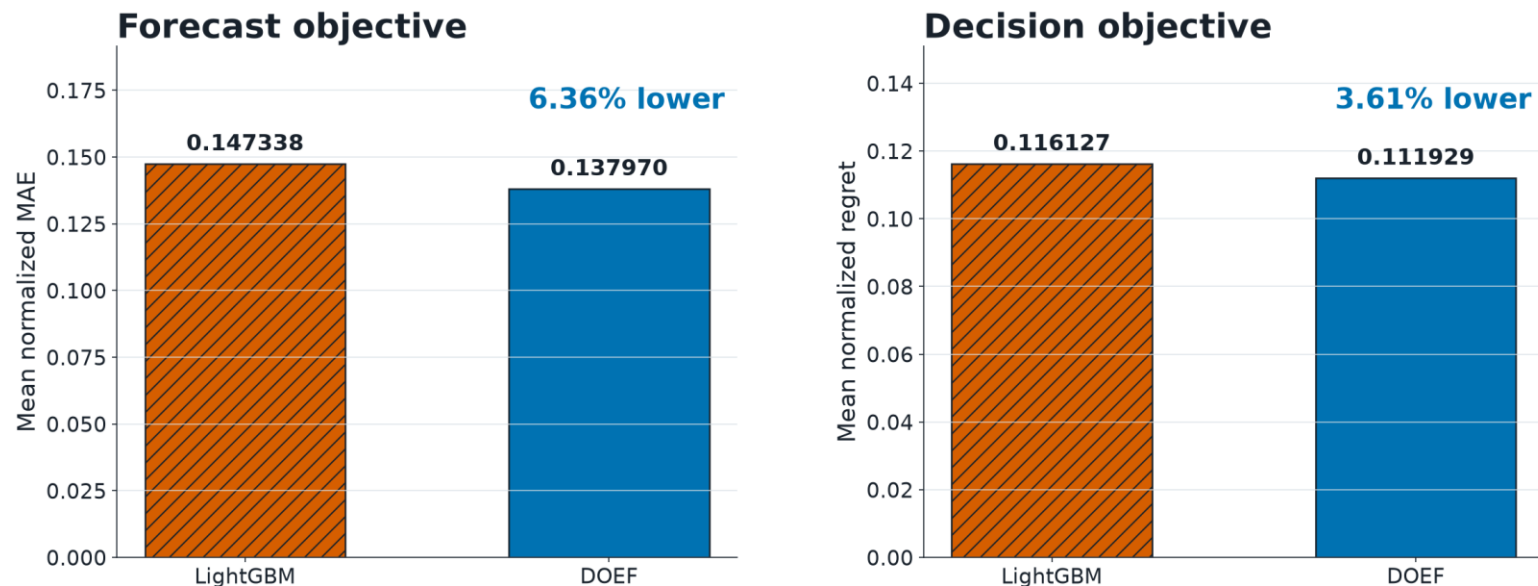
Frozen method visual: [outputs/phase13_1/architecture/doef_architecture_master.png](#) · formula: $w_b \cdot \text{LightGBM} + (1-w_b) \cdot \text{DayWeek}$

预测误差降低 6.36%，决策后悔值降低 3.61%

相对 LightGBM，八建筑等权平均的归一化 MAE 和归一化决策后悔值均下降。

DOEF improves both frozen aggregate objectives

Frozen eight-building Test aggregate · comparison against LightGBM · lower is better



Building-level decision regret: 6 wins · 2 ties · 0 losses

Peak-shaving context: mean 6.44%; minimum -1.02% (negative case retained). Oracle-relative regret is not monetary regret.

6.36%

归一化 MAE 相对改善

3.61%

归一化后悔值相对改善

6 / 2 / 0

win / tie / loss

最小整段削峰率: -1.017%

预测误差与决策后悔值

RMSE 衡量负荷预测偏差；决策后悔值衡量相对 Oracle 的实际峰值差。

预测指标

RMSE

预测负荷与实际负荷的偏差

对各时刻的预测误差平方取平均，再开方。
用于评价负荷预测质量。

调度指标

regret

相对 Oracle 的实际峰值差

预测生成电池计划，用实际负荷评价峰值。
Oracle 知道未来负荷，仅作事后参照。

双层评价



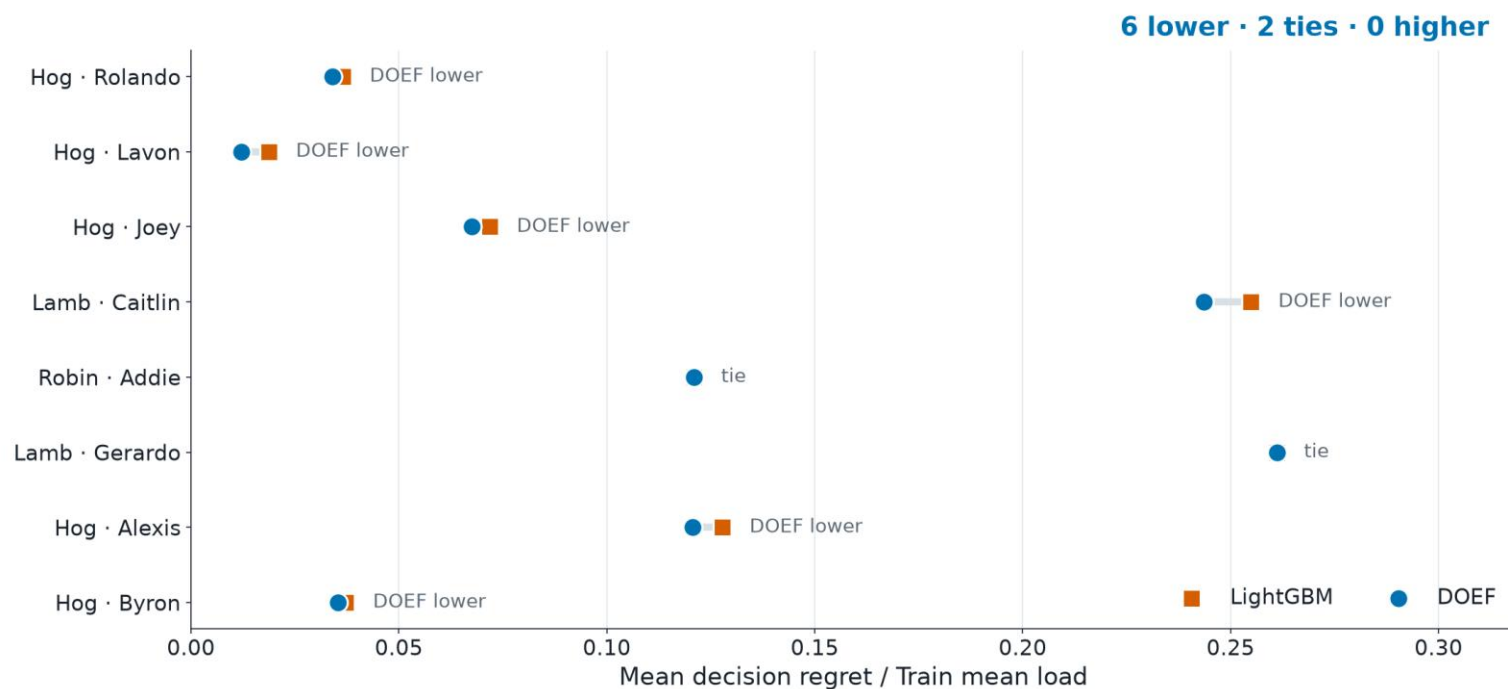
同时报告预测精度与调度表现

八建筑后悔值对比：6 胜、2 平、0 负

296 栋办公建筑候选中，82 栋通过质量筛选，再依据训练期负荷形态选出 8 栋。

Decision value across all eight fixed buildings

Frozen Test mean decision regret normalized by each building's frozen Train mean load · lower is better



All eight preselected offices are shown in frozen selection order. This is multi-building robustness, not unseen-building transfer.

Frozen source: SCI-03 · buildings preserved in preselected frozen order; no sorting/deletion

8 / 8

全部展示

6

lower

2

ties

0

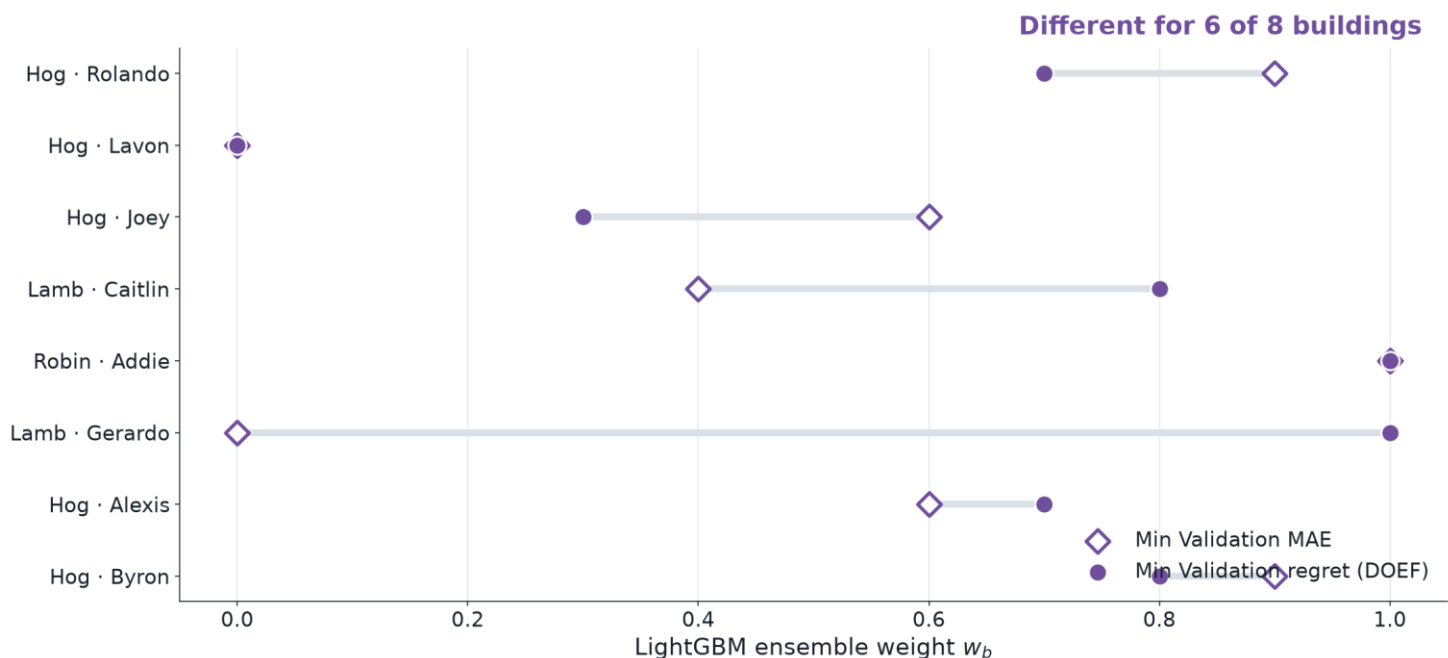
higher

验证集权重选择与测试集评价

验证集比较 11 个候选权重；测试集只用于评价，不参与模型、权重或平局规则选择。

Validation objectives select different ensemble weights

Per-building LightGBM weight; remaining weight is DayWeek · selection uses Validation only



Test is excluded from weight and tie-break selection. Points show frozen Phase 8 weights; no optimization was rerun.

验证后固定权重

Validation

搜索 $0.0 \rightarrow 1.0$
步长 0.1



FREEZE w_b

Test

evaluation only

Frozen source: SCI-04 · candidate grid = 11 weights · prior historical Test visibility means "pristine blind test" is not claimed

预测与储能调度的仿真流程

离线仿真已实现预测、优化和实际负荷评价；真实楼宇在线部署仍需验证。



REPEATABLE

FROZEN

NOT DEPLOYED

已实现负荷预测、约束调度和实际峰值评价的仿真流程。

决策导向融合与多建筑评测

通过验证集调度表现选择预测组合，并在相同电池约束下开展多建筑比较。

Four verified contribution pillars

01 Causal forecasting

24 h horizon
positive lags + shifted history
no future-target leakage

02 Decision-oriented ensemble

Validation regret selects w_b
LightGBM + DayWeek
Test excluded from selection

03 Multi-building robustness

8 fixed heterogeneous offices
Train-only morphology sampling
not unseen-building transfer

04 Storage decision evaluation

same constrained optimizer
realized peaks + Oracle regret
alongside forecast metrics

CONCEPTUAL / EXPLANATORY · contribution is project-scoped; no first / SOTA / end-to-end claim

Conceptual asset: CON-02 · no first/world-first/SOTA/end-to-end deployment claim

面向储能削峰的 负荷预测组合选择

FOR BATTERY PEAK SHAVING

用验证集调度表现选择负荷预测组合，在本次八建筑实验中取得改善。

DOEF v1.0 · 八建筑仿真验证

8 栋办公建筑 · 单一测试月 · 24 h 超前预测 · 离线仿真

01 预测误差 $\neq$ 决策价值

RMSE 与 regret 回答不同问题

02 验证集选择权重

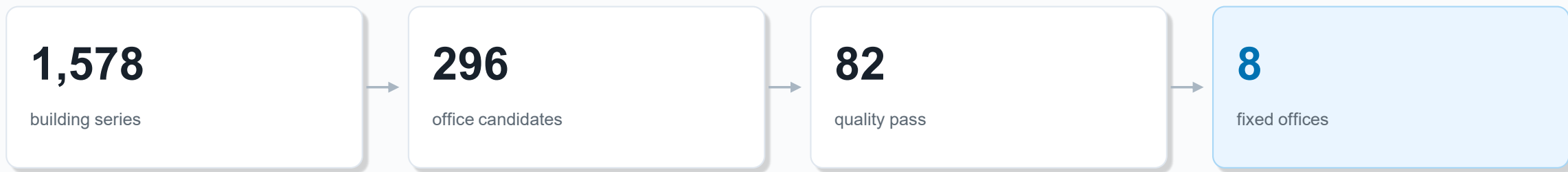
按决策后悔值确定 w_{β}

03 测试集评价效果

8 栋建筑分别训练与评价

建筑筛选与样本构成

先按数据质量确定候选范围，再用训练期负荷形态选楼；不使用验证或测试成绩。



建筑样本

01 Hog · Rolando

05 Robin · Addie

02 Hog · Lavon

06 Lamb · Gerardo

03 Hog · Joey

07 Hog · Alexis

04 Lamb · Caitlin

08 Hog · Byron

Train-only morphology features

- log mean load
- peak-to-average ratio
- lag-24 autocorrelation
- weekday-weekend difference
- robust outlier fraction
- coefficient of variation
- peak-valley range
- lag-168 autocorrelation
- daily-profile variability

NO Validation/Test performance used

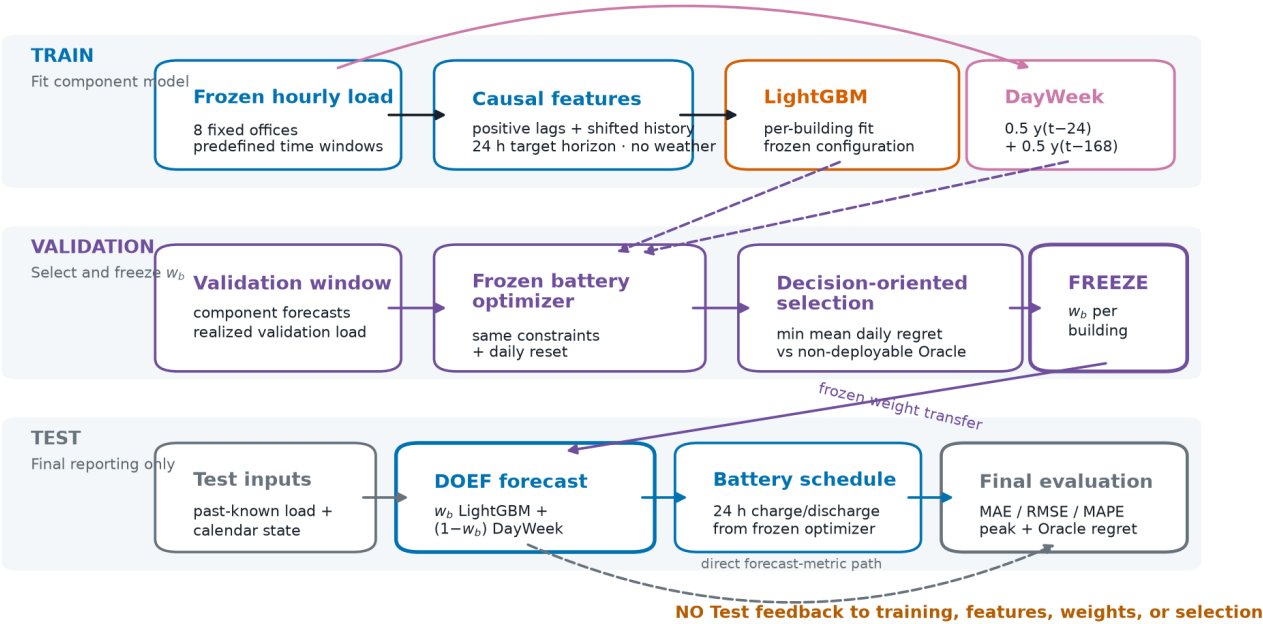
Frozen source: outputs/phase7/selected_buildings.json

因果特征与时间切分

24 h 超前预测仅使用当时可获得的负荷和日历信息，按目标可用性处理切分边界。

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信息边界

- positive lags
- shifted history
- calendar state
- no weather
- no target leakage

历史测试期已被查看

Frozen source: ARCH-01 · prior Test visibility caveat retained

预测基线与模型组成

DayWeek 与逐建筑 LightGBM 构成 DOEF，其余模型用于基线比较。

Method	Definition / input	Role in benchmark	Novelty claim?
Day Persistence	t-24 h	Baseline	No
Week	t-168 h	Baseline	No
DayWeek	0.5·Day + 0.5·Week	DOEF component	No
LightGBM	causal features	DOEF component / baseline	No
XGBoost	causal features	Comparator	No
CatBoost	causal features	Comparator	No
DOEF	$w_{\beta} \cdot \text{LGBM} + (1 - w_{\beta}) \cdot \text{DayWeek}$	Final frozen method	Framework
Peak-aware LGBM	supporting experiment	Rejected support	No

Frozen benchmark definitions · no model-family catalogue in main deck

电池参数与调度约束

各预测方法共享同一建筑的电池与优化器，电池规模由训练期负荷确定。

Capacity

10% of mean daily energy

kWh

Pcharge / Pdischarge

25% of capacity per hour

kW

SOC range

10% → 90%

kWh state

Initial / terminal SOC

50% / 50%

daily reset

η charge / η discharge

0.94868 / 0.94868

round-trip 0.90

Optimizer

deterministic daily peak shaving

scipy.optimize.milp

1-hour interval: load energy kWh • dispatch power kW • capacity/SOC kWh

Frozen source: outputs/phase7/building_battery_configs.csv · same battery/optimizer for all forecast methods

八建筑预测结果

八建筑测试集等权平均；MAE、RMSE 和后悔值按各建筑训练期平均负荷归一化。

Method	normalized MAE	normalized RMSE	normalized regret	peak reduction mean
Day Persistence	.225885	.430193	.162986	-1.878%
Week	.201578	.328414	.144323	3.085%
DayWeek	.183478	.296621	.143019	4.907%
LightGBM	.147338	.216942	.116127	1.008%
XGBoost	.149781	.216751	.117534	0.453%
CatBoost	.173604	.239211	.118740	-24.482%
DOEF	.137970	.207822	.111929	6.442%
Peak-aware LGBM	.147587	.217284	.116202	1.129%

Canonical source: outputs/phase9/final_benchmark.csv · lower is better for normalized MAE/RMSE/regret

削峰率分布与失效案例

均值、中位数和最小值共同描述调度表现，负削峰率表示取电峰值反而升高。

均值与中位数的差异

CatBoost peak reduction

-24.48% +4.82%

mean median

极端负值拉低平均削峰率，需结合中位数和最小值判断调度表现。

min = -245.94%

Method	norm regret	peak mean	peak median	peak min
LightGBM	.116127	1.008%	5.057%	-41.192%
XGBoost	.117534	0.453%	4.588%	-44.357%
CatBoost	.118740	-24.482%	4.821%	-245.943%
DOEF	.111929	6.442%	5.629%	-1.017%

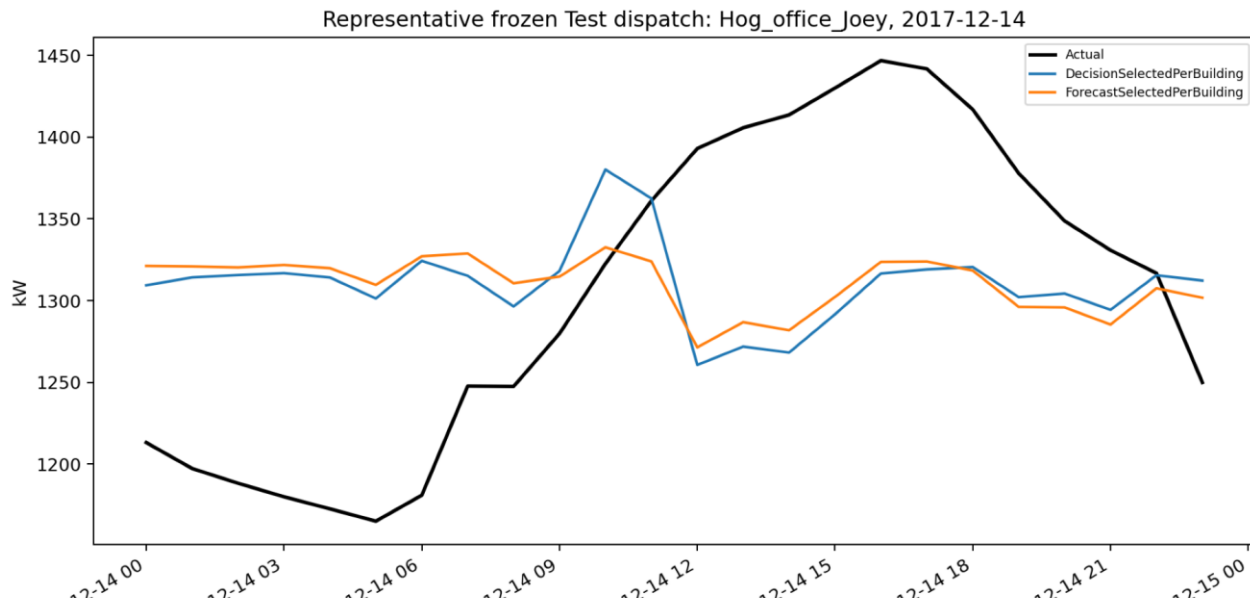
Frozen source: outputs/phase9/final_benchmark.csv + decision metrics · peak reduction is not a universal scalar

预测误差对充放电计划的影响

预先选定的测试案例用于解释预测如何改变调度结果，不能代表全部建筑表现。

Frozen representative forecast-to-dispatch trace

Phase 8 pre-frozen Test case · unchanged raster evidence · presentation frame only



Read as a mechanism trace, not a universal performance claim; the case and values remain frozen.

案例说明

- 预先确定的测试案例
- 用于说明调度机制
- 展示预测与实际负荷
- 关注高负荷时段的偏差
- 总体表现见全样本结果

Exact frozen source: outputs/phase13_1/figures/sci_05_storage_dispatch_frozen.png · no value extraction or case reselection

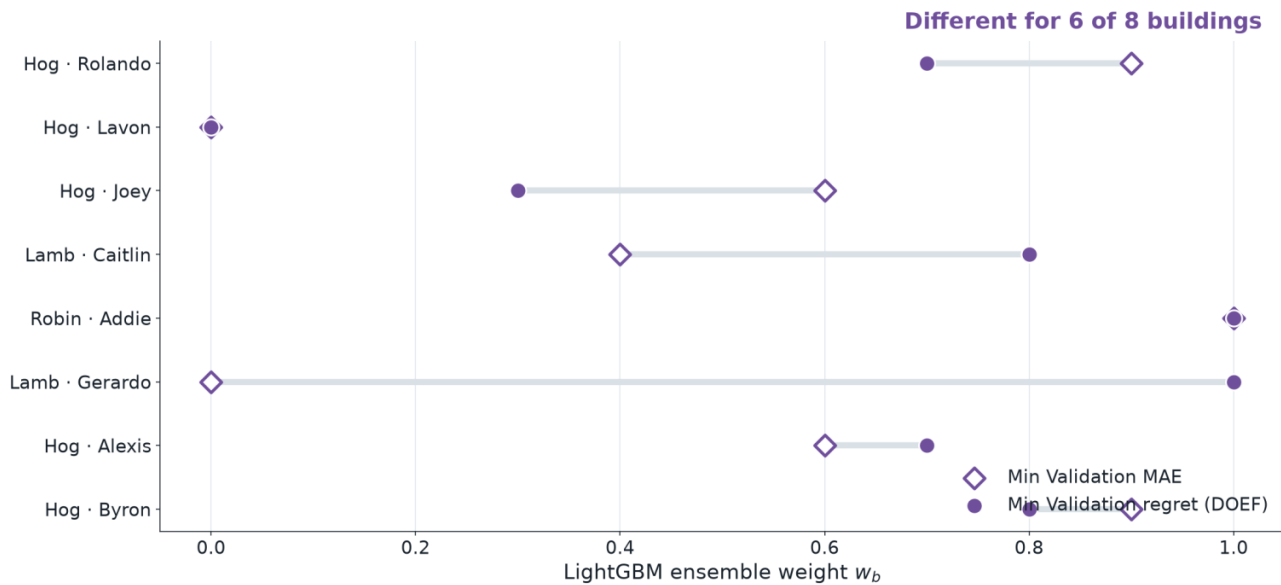
逐建筑融合权重

候选权重为 0.0、0.1、...、1.0，按验证集目标和平局规则选择，测试期保持不变。

Joey: 0.6 → 0.3

Validation objectives select different ensemble weights

Per-building LightGBM weight; remaining weight is DayWeek · selection uses Validation only



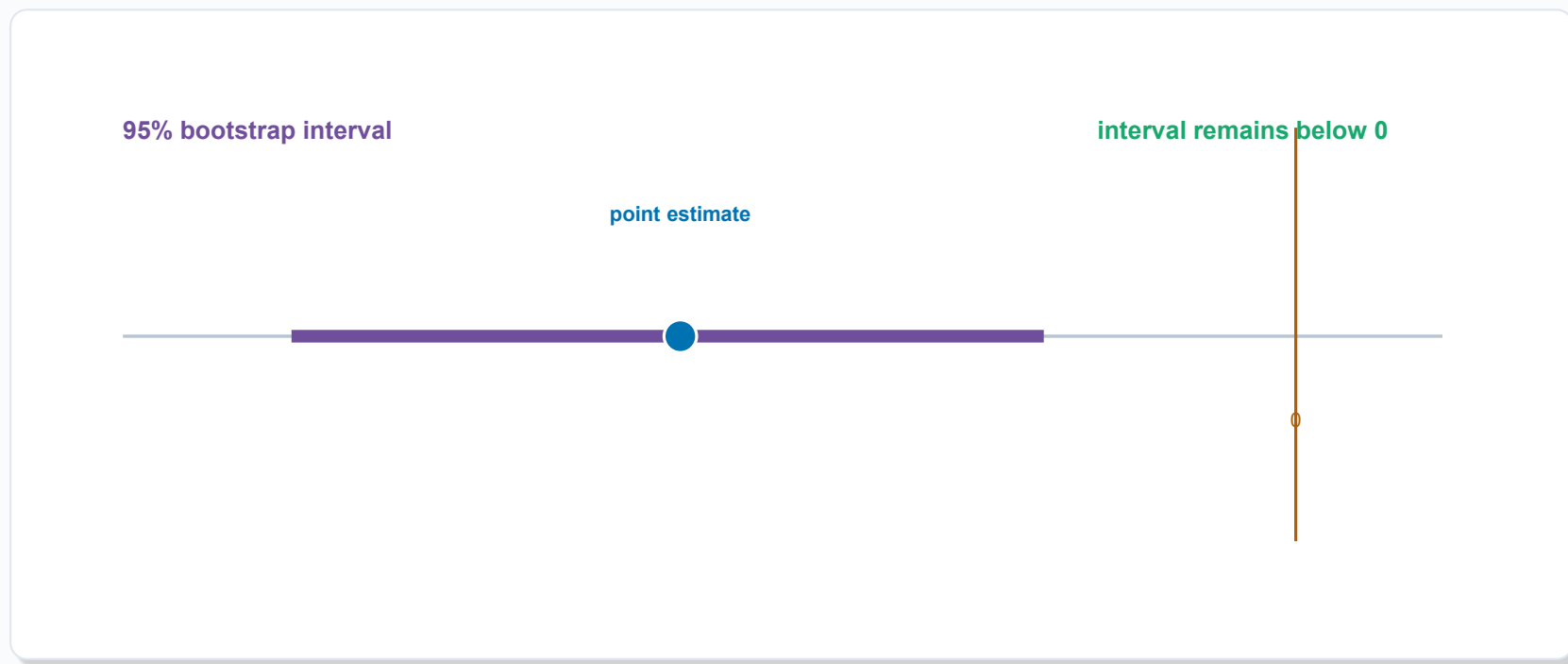
Test is excluded from weight and tie-break selection. Points show frozen Phase 8 weights; no optimization was rerun.

Building	w forecast	w decision
Rolando	.9	.7
Lavon	0	0
Joey	.6	.3
Caitlin	.4	.8
Addie	1	1
Gerardo	0	1
Alexis	.6	.7
Byron	.9	.8

Frozen source: selected_weights.csv + SCI-04 · no interpolation between grid points

建筑级 bootstrap 不确定性分析

以建筑为单位进行配对重采样，区间描述本次 8 栋建筑样本内的不确定性。



95% interval

[-0.006850 , -0.001719]

10,000

resamples · seed 42

-0.004198

point estimate

fixed 8-building scope only

Frozen source: outputs/phase9_1/final_reporting_summary.json · no new resampling

DOEF 公式与预测输出的一致性

按公式复算 5,760 个测试预测，与保存的输出逐点比对；该检查不涉及重新训练。

$$w_{\beta} \times \text{LightGBM} + (1 - w_{\beta}) \times \text{DayWeek}$$

=

保存的融合预测
ensemble_prediction

逐点一致性检查

5,760

个预测值参与比对

0.0 kW

max absolute difference

8 buildings × 720 Test hours

公式复算与保存的预测一致
容差 $1e-12$

Frozen source: Phase 9.1 reporting summary · identity check only

稳健调度的对比实验

验证集选出的稳健调度方案在测试期后悔值更高；保留该结果，未反调参数。

Validation selection

robust $\lambda = 0.5$

λ 由验证集选择；测试期保持不变。

selection integrity

Test mean regret

20.4635 kW

robust variant

11.1274 kW

deterministic

该稳健方案的测试后悔值高于确定性对照，
最终 DOEF 未采用这一方案。

稳健调度失效案例